

Two pointer (Number of unique elements in an sorted array)

↳ Sorted array means duplicates are always adjacent - compare current element with previous to detect unique element

arr $\rightarrow$

1	1	2	3
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 ↑
 j

if (arr[j] == arr[j-1]) $\rightarrow$ they are not unique

↳ Two pointer works together, one scans (j), one tracks the write position (i)

↳ First element is guaranteed unique, so i=0, j=1, unique count starts at 1.

↳ arr[j] == arr[j-1] $\rightarrow$ duplicate, skip
arr[j] != arr[j-1] $\rightarrow$ unique, place at i+1
 i++
 unique count ++

↳ Time complexity
O(n)

Space complexity
O(1) $\rightarrow$ since we used no extra space

```
while (j < len)
{
    if (arr[j] == arr[j-1])
    {
        j++;
        continue;
    }
    unique-count++;
    i++;
    j++;
}
```

on starting
unique-count = 1
↓
because first
element is always
unique